

Higher-order Topological Type-II Hyperbolic Lattices

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Supplementary Note 1 Construction of type-II hyperbolic lattice and the corresponding Bravais lattice

Supplementary Fig. 1a illustrates an infinite $\{8, 3\}$ type-I hyperbolic lattice (grey lines) together with its corresponding $\{8, 8\}$ Bravais lattice (green lines) whose unit cell is highlighted as a green octagon. The type-II lattice considered in this work is generated from the regular type-I hyperbolic lattice through the conformal mapping:

$$w(z) = e^{-\frac{\pi}{2r_h} \left(\frac{1+z}{1-z} \right)^{\frac{i}{r_h}}} \quad (\text{S1})$$

Under this mapping, the type-I hyperbolic lattice is transformed into the type-II geometry $\{r_h, 8, 3\}$, as illustrated in Supplementary Fig. 1b. At the same time, the corresponding type-I Bravais lattice is transformed into a type-II Bravais lattice $\{r_h, 8, 8\}$ type-II Bravais lattice, , thereby redefining the unit-cell structure in the transformed geometry. This construction shows that the type-II hyperbolic lattice is not an arbitrary finite cluster, but is instead generated from a regular hyperbolic lattice through a conformal transformation

Supplementary Note 2: Symmetry analysis of the higher-order topological phase

In this note, we summarize the symmetry properties of the higher-order Hamiltonian H_{HOTI} and present the corresponding disorder test for the zero-energy corner states.

After the Wilson mass term is introduced, the higher-order Hamiltonian no longer preserves time-reversal symmetry. Nevertheless, it still preserves the particle-hole symmetry $P = \sigma_3 s_1 K$ and the composite symmetry Sm_z , where $m_z = \sigma_3 s_0$ represents the mirror symmetry operator, $S = PT$ is the chiral symmetry operator. In addition, for the rotationally symmetric hyperbolic lattice, the combined symmetries $C_p T$ and $C_p m_z$ are also present. Therefore, after the Wilson mass term breaks time-reversal symmetry, the zero-energy corner states remain symmetry protected.

For the disorder analysis considered here, the most relevant protecting symmetries are the particle-hole symmetry P and the effective chiral symmetry Sm_z . The reason is that random onsite disorder generally breaks the rotation-related combined symmetries $C_p T$ and $C_p m_z$, whereas the preservation or breaking of P and Sm_z directly determines whether the zero-energy corner states remain robust against weak disorder.

To verify this protection mechanism, we consider two types of onsite disorder terms,

$$\Delta H_1 = W_1 \sum_{\langle i \rangle} n_i c_i^\dagger \sigma_2 s_3 c_i, \quad \Delta H_2 = W_2 \sum_{\langle i \rangle} n_i c_i^\dagger \sigma_1 s_3 c_i \quad (\text{S2})$$

Where n_i is uniformly distributed in $[-1, 1]$. These two disorder terms are chosen according to their symmetry properties. The first preserves both the particle-hole symmetry P and the effective chiral symmetry Sm_z , whereas the second breaks both. The corresponding energy spectra and quadrupole moments are shown in Supplementary Fig. 2. As shown in Supplementary Fig. 2(a) and 2(b), under weak symmetry-preserving disorder, the eight degenerate zero-energy corner states remain stable and the quadrupole moment stays quantized at 0.5 until the disorder strength exceeds a critical value $W_1 \approx 0.78$. By contrast, as shown in Supplementary Fig. 2(c) and 2(d), under weak symmetry-breaking disorder, the degeneracy of the zero-energy states is lifted immediately, and the quadrupole moment deviates from the quantized value. These results provide direct numerical evidence that the higher-order corner states in the type-II hyperbolic lattice are protected by particle-hole symmetry P and the effective chiral symmetry Sm_z .

Supplementary Note 3: Calculation of the generalized quadrupole moment

For the type-II hyperbolic lattice with corner states larger than 8, the conventional real-space quadrupole moment cannot be directly applied in its original form. To characterize such cases, a generalized quadrupole construction is introduced based on an explicit coordinate transformation adapted to the corner-state distribution.

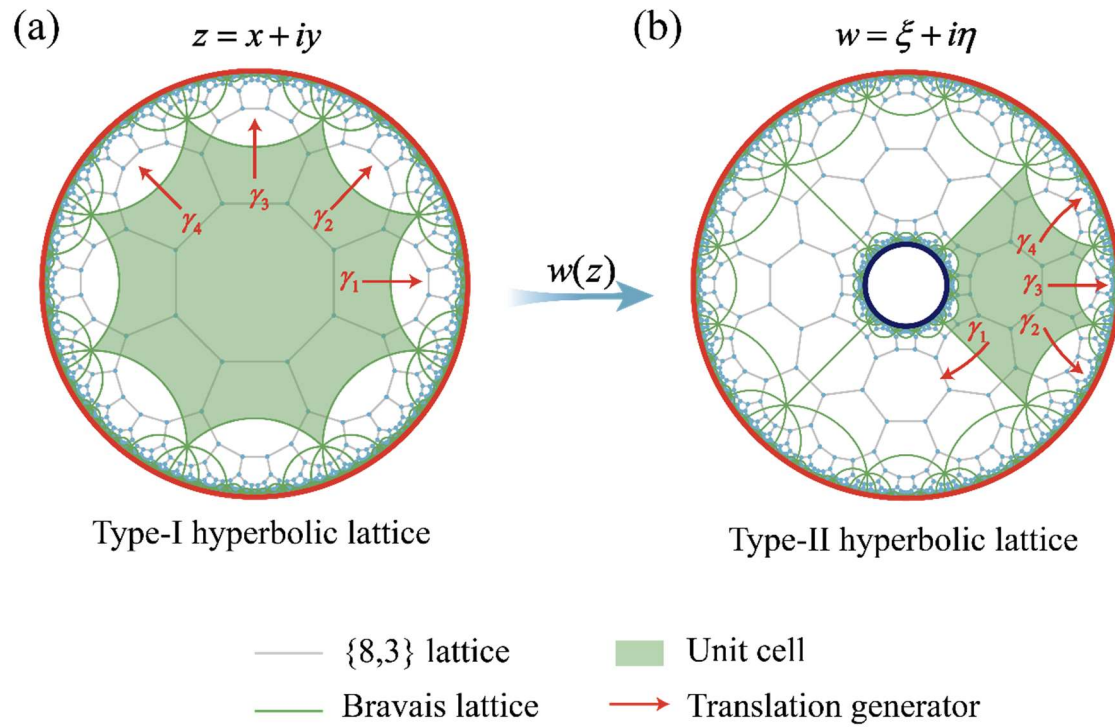
Here we consider the $k = 12$ type-II hyperbolic lattice, for which the system hosts twenty-four corner states. As shown in Supplementary Fig. 3(a), the corner states are distributed over both the inner and outer boundaries. The full boundary is partitioned into four sectors, and each sector contains six corner states in total, with three on the inner boundary and three on the outer boundary.

To construct the generalized quadrupole moment, a sector-wise coordinate transformation is performed so that the corner-state distribution is mapped into an effective geometry suitable for quadrupole characterization, as illustrated schematically in Supplementary Fig. 3(b). If the coordinate of a site n is written in the form $r_n e^{i\theta_n}$

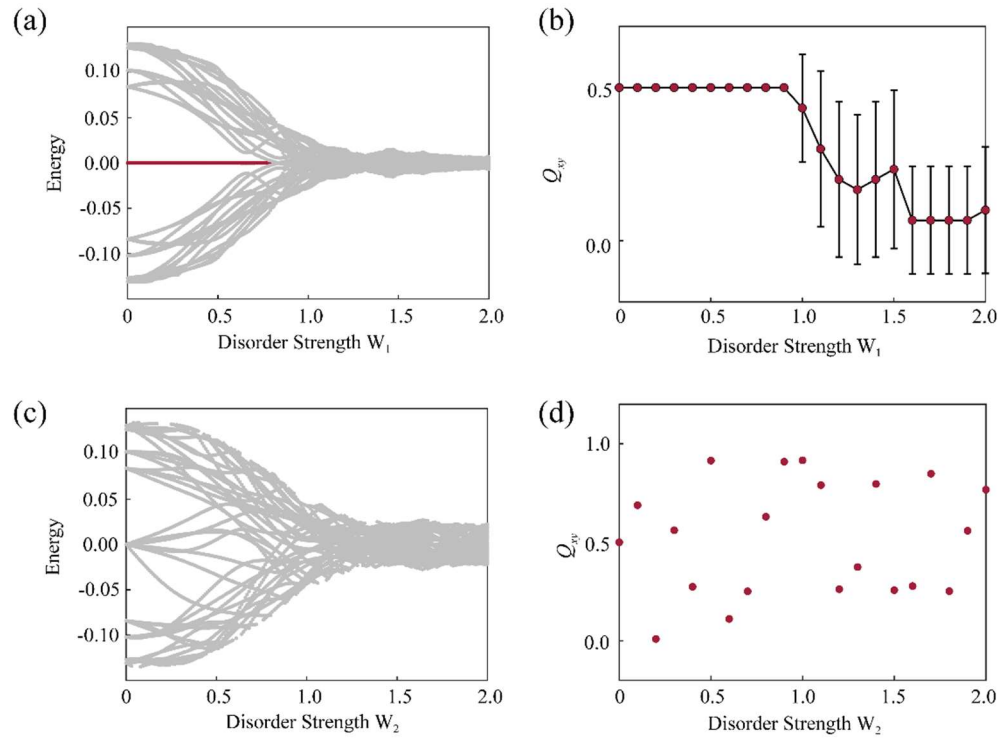
The transformed coordinate $r_n e^{i\theta'_n}$ is defined piecewise as follows: for $0 \leq \theta_n <$

$5\pi/6$, $\theta'_n = 3\theta_n/5$; for $5\pi/6 \leq \theta_n < \pi$, $\theta'_n = 3\theta_n - 2\pi$; for $-\pi \leq \theta_n < -\pi/6$, $\theta'_n = 3\theta_n/5 - 2\pi/5$; and for $-\pi/6 \leq \theta_n < 0$, $\theta'_n = 3\theta_n$.

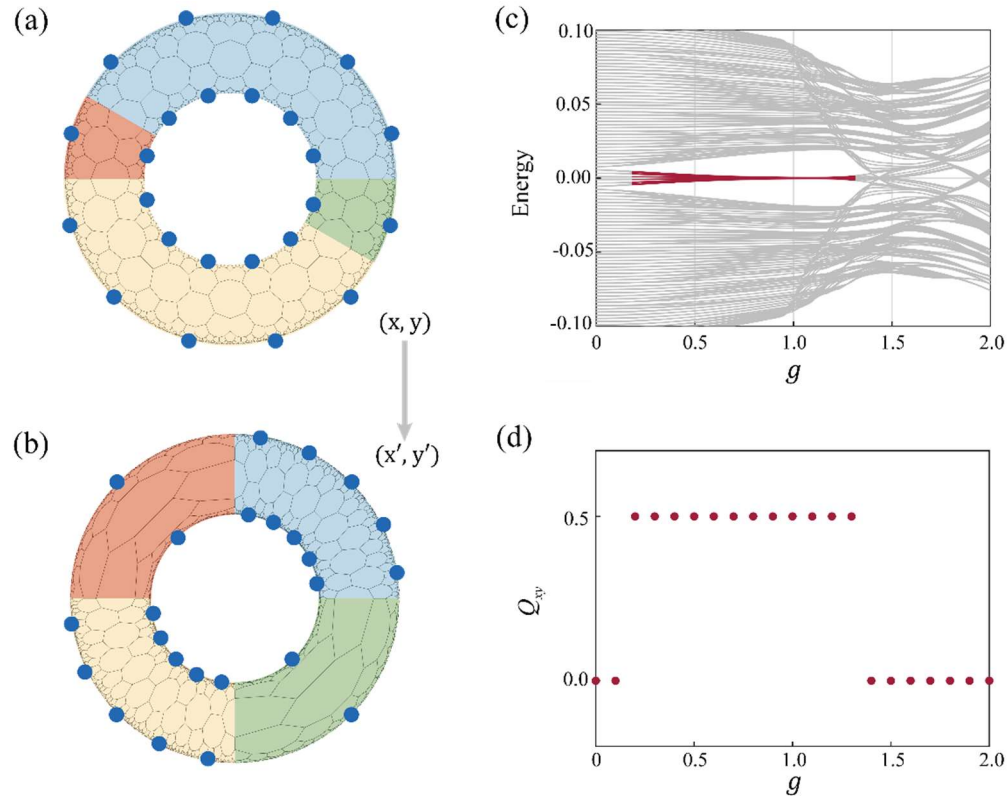
Using this transformed coordinate representation, the generalized quadrupole moment is computed and compared with the energy spectrum of the higher-order Hamiltonian as a function of the Wilson mass. The results are shown in Supplementary Fig. 3(c) and Supplementary Fig. 3(d). It is found that the generalized quadrupole moment $Q_{x'y'}$ remains quantized at 0.5 for finite g , indicating a non-trivial topological phase. When g exceed 1.3, the system enters a topological trivial phase. This provides additional evidence that the higher-order type-II lattice remains topological nontrivial even when the number of corner states increases to twenty-four. The same construction can also be applied to the $k = 8$ case with sixteen corner states, leading to the same qualitative conclusion.



Supplementary Figure 1 | Conformal generation of the type-II hyperbolic lattice from its type-I counterpart. (a) Infinite $\{8, 3\}$ type-I hyperbolic lattice (gray lines) and its corresponding $\{8, 8\}$ Bravais lattice (green lines), with one unit cell highlighted. (b) Infinite type-II hyperbolic lattice obtained from the type-I lattice through the conformal mapping $w(z)$ described in the text, together with the corresponding transformed Bravais-lattice construction.



Supplementary Figure 2 | The symmetry protection of the zero-energy corner states under onsite disorder. (a) Energy spectrum versus disorder strength W_1 for the symmetry-preserving disorder term ΔH_1 at characteristic radius $r_h = 0.365, k = 4, g = 0.5$. The zero-energy corner states are marked in red. (b) Quadrupole moment Q_{xy} as a function of W_1 for the symmetry-preserving disorder term. (c) Energy spectrum versus disorder strength W_2 for the symmetry-breaking disorder term ΔH_2 (d) Quadrupole moment Q_{xy} as a function of W_2 for the symmetry-breaking disorder term.



Supplementary Figure 3 | Generalized quadrupole moment characterization for the type-II hyperbolic lattice with twenty-four corner states. (a) Schematic of the type-II hyperbolic lattice $\{0.715, 8, 3\}$, in which the corner states are distributed over the inner and outer boundaries. The boundary is partitioned into four sectors, each sector contains six corner states in total. (b) Schematic illustration of the coordinate transformation used to map the sector-wise corner-state distribution into an effective geometry suitable for defining generalized quadrupole moment. (c) Energy spectrum of the higher-order Hamiltonian as a function of the Wilson mass for the lattice shown in (a). (d) Corresponding generalized quadrupole moment $Q_{x'y'}$ as a function of g .